

Exhaustive Search

* It is a problem-solving method that systematically tries all possible candidate solutions & selects the one that satisfies the given problem constraints & produces the optimal result.

* Examples

- Travelling Salesperson Problem (TSP)
- Knapsack problem
- Job assignment problem, etc.

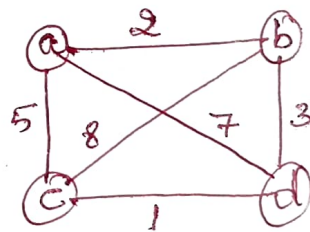
Travelling Salesperson problem

* problem statement

To find the shortest tour through a given set of 'n' cities that visits each city exactly once before returning to city where it started.

problem can be stated as the problem of finding the shortest Hamiltonian circuit of the graph.

* Example



Tour

Length

1) $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$

$2 + 8 + 1 + 2 = 18$ } ✓

2) $a \rightarrow d \rightarrow c \rightarrow b \rightarrow a$

$7 + 1 + 8 + 2 = 18$ }

3) $a \rightarrow b \rightarrow d \rightarrow c \rightarrow a$

$2 + 3 + 1 + 5 = 11$ } optimal

4) $a \rightarrow c \rightarrow d \rightarrow b \rightarrow a$

$5 + 1 + 3 + 2 = 11$ }

5) $a \rightarrow c \rightarrow b \rightarrow d \rightarrow a$

$5 + 8 + 3 + 7 = 23$

6) $a \rightarrow d \rightarrow b \rightarrow c \rightarrow a$

$7 + 3 + 8 + 5 = 23$

Shortest tour = $a \rightarrow b \rightarrow d \rightarrow c \rightarrow a$ (OR) $a \rightarrow c \rightarrow d \rightarrow b \rightarrow a$